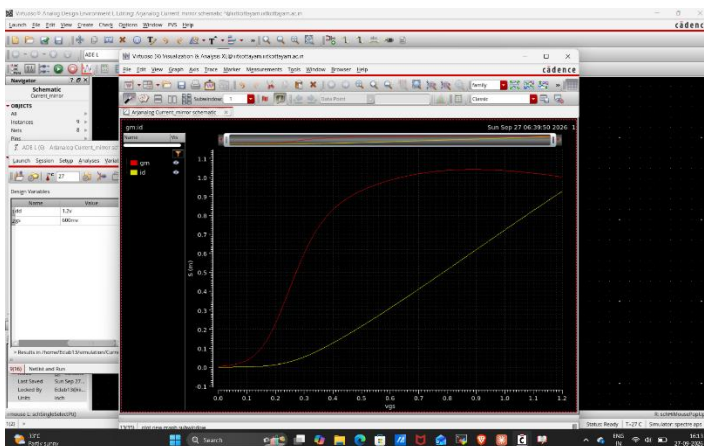
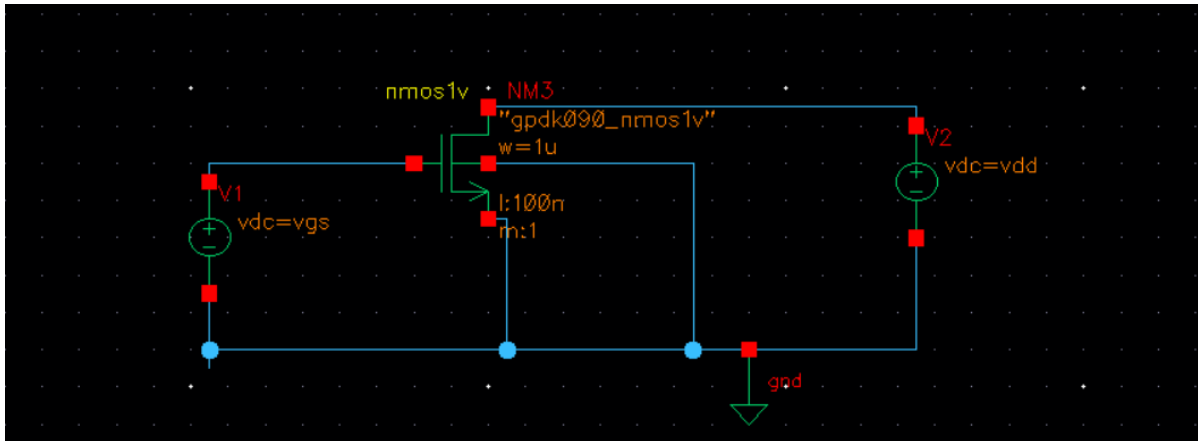


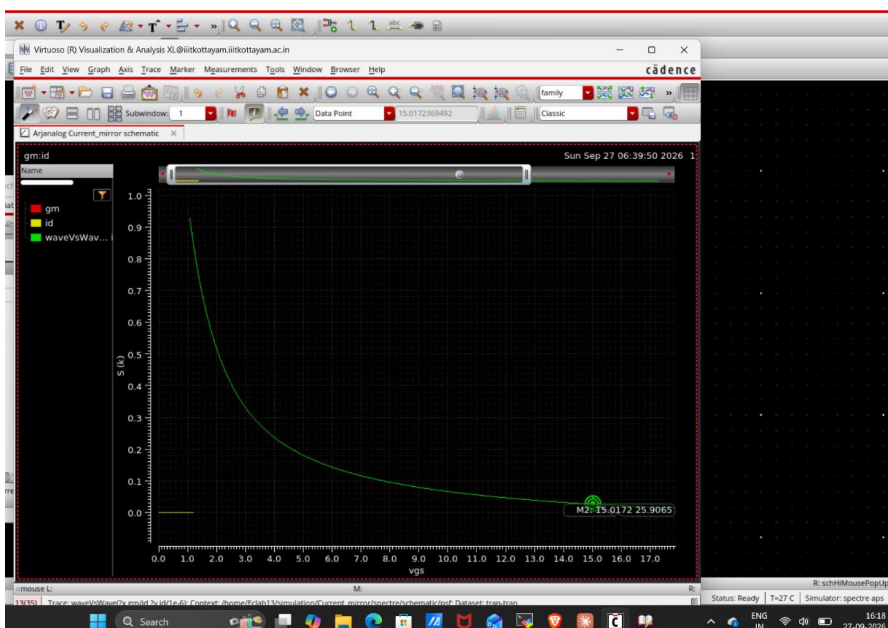
Click here for git hub access https://github.com/ArjunRaj30/AnalogElectronicsLab_2025BEC0025

To design simple current mirror and cascode current mirror for 1 mA output current. Take g_m/I_d of 15 for sizing. Determine size of transistor to produce a current of 1mA. (use g_m/I_d characterization done in lab for sizing). Design cascode current mirror as well.

NMOS Characterisation -



Gm/Id vs Id/1um graph



From Reference Nmos

X value (g_m/I_D): $15.018V^{-1}$

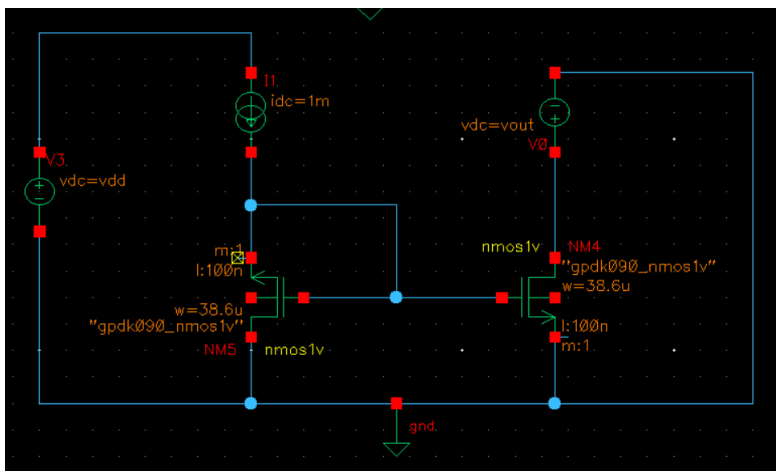
Y value (I_D/W): $25.9018\mu A/\mu m$

$\therefore$ Required W [for $1/3(g_m/I_D) = 15$]

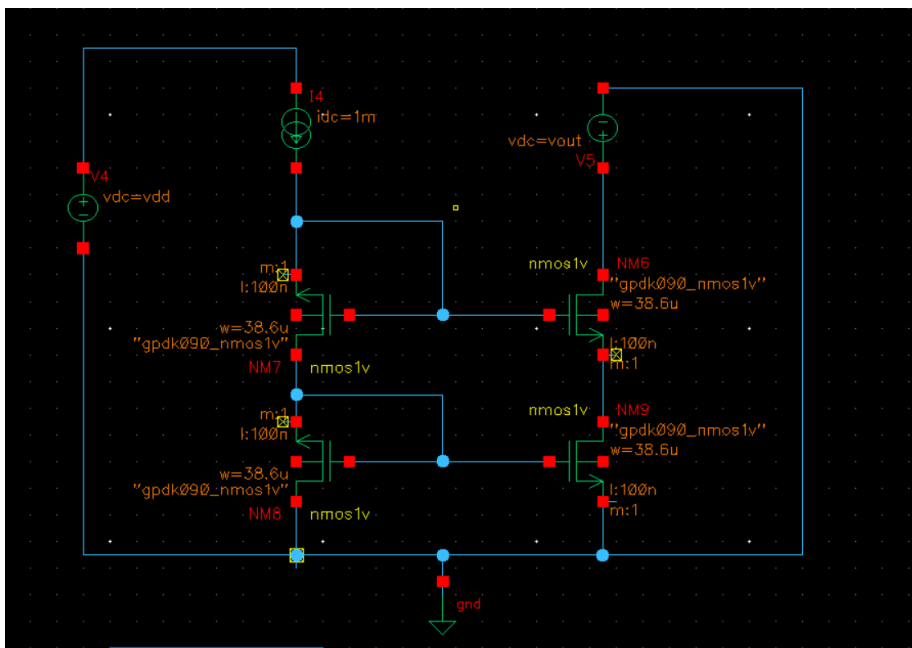
$\Rightarrow \frac{1mA}{0.0259018mA/\mu m} \approx \underline{\underline{38.6\mu m}}$

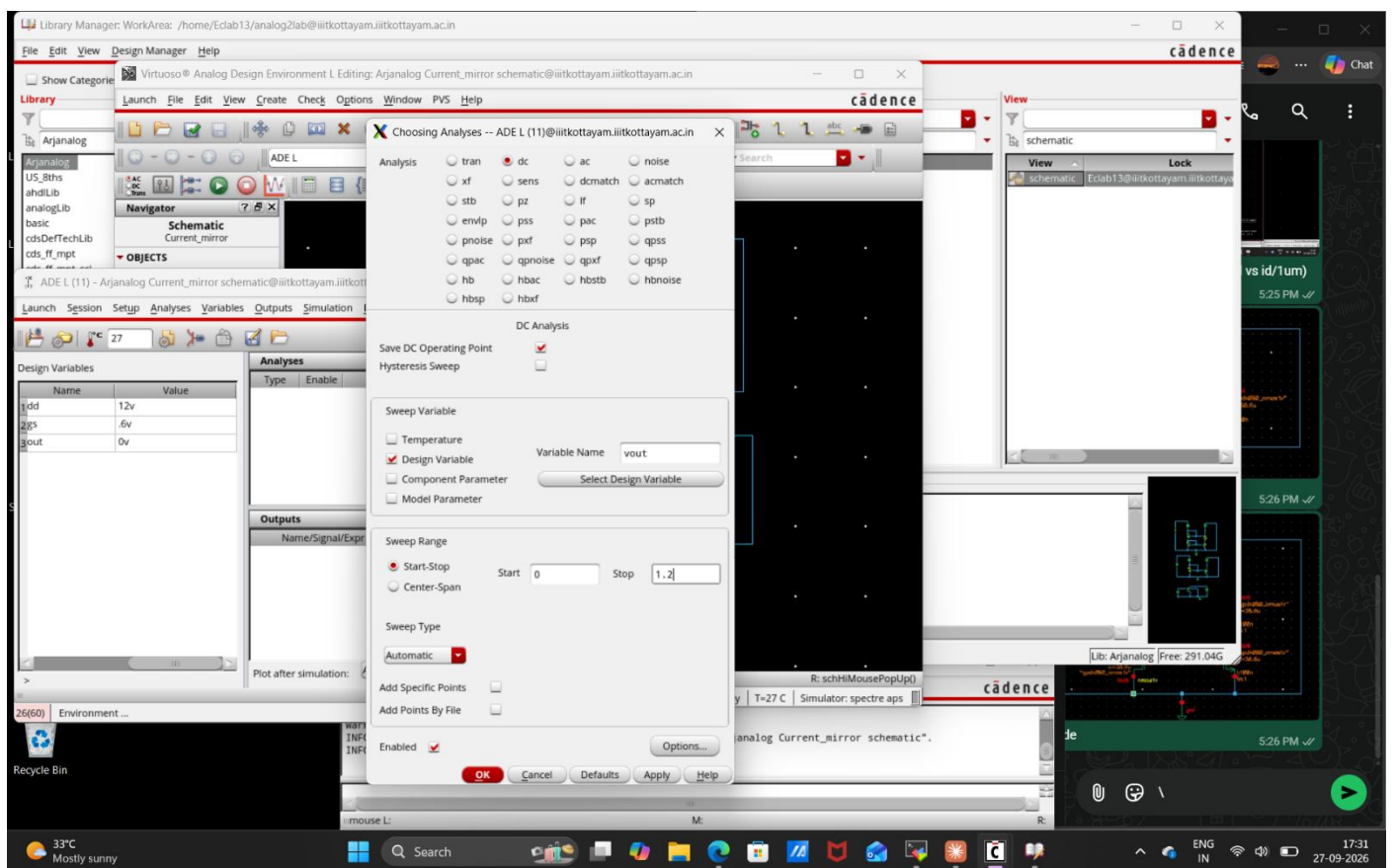
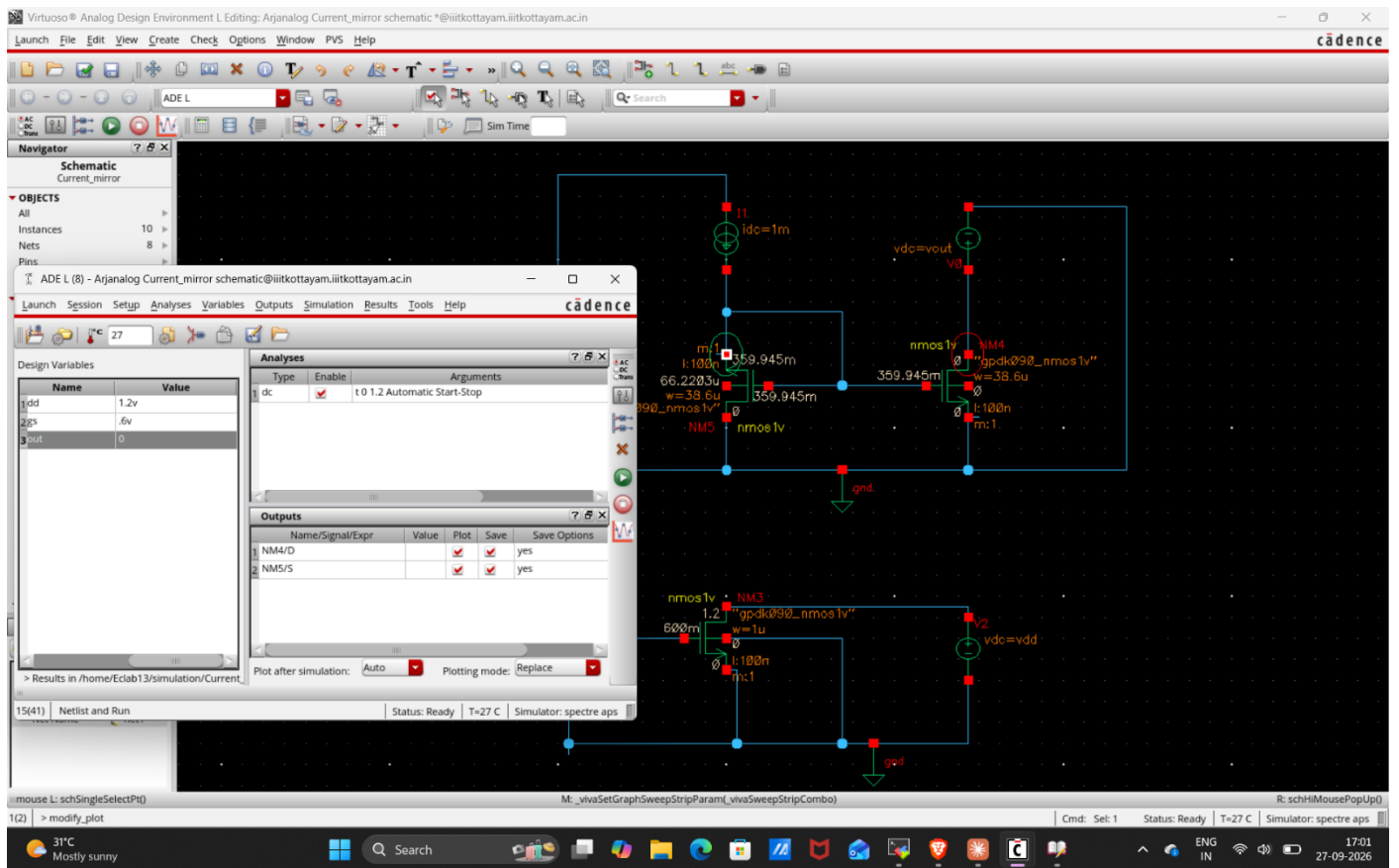
L is fixed as 100nm

Simple Current Mirror



Cascode Current Mirror

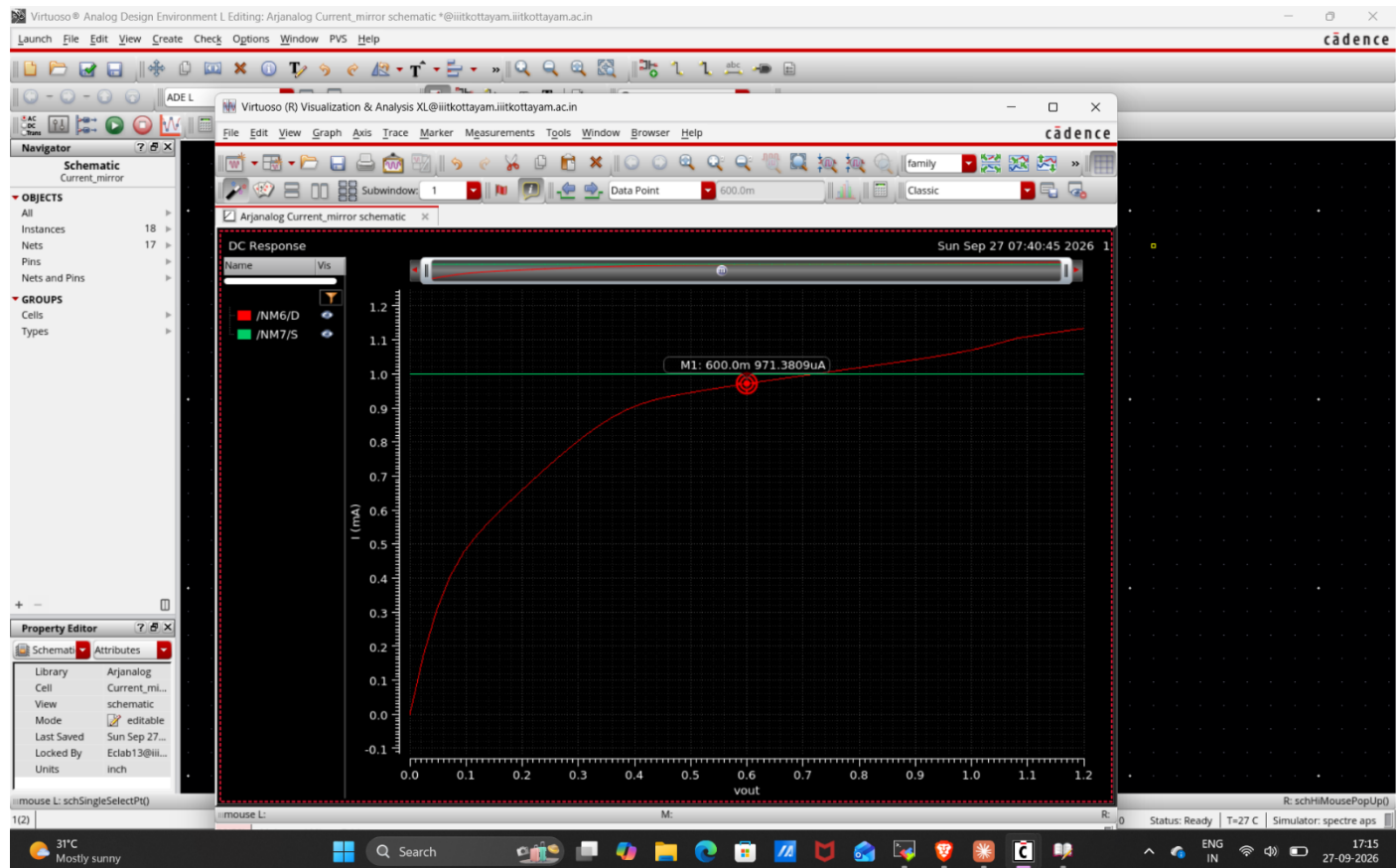




Simple current mirror lin and lout graph



Cascode Current mirror lin vs lout graph



Simple and Cascode current mirror lout comparison and $V_{gs} = V_{dd}/2$ ie .6v



To make $I_{out} = 2mA$
 $\rightarrow \frac{I_{dref}}{W/L_{ref}} = \frac{I_{dout}}{W/L}$
 W/L of 2nd mosfet of Simple current mirror becomes $(W/L) \times 1/2$ ie
 $W \rightarrow W/2$ ie 19.3 μm
 Both NM6 and NM9 $\rightarrow$ 19.3 μm width in Cascode

Inference

The cascode mirror shows flatter I_{out} vs v_{ds} response (higher output resistance) than the simple mirror, at the cost of a larger minimum output voltage (compliance voltage) needed to keep all devices in saturation.

Therefore for precision and high output resistance, the cascode is way better.

For low supply voltage operation, the simple mirror is more practical.